Boris's Brexit Dilemma as a Game — class version

Lecture 3 case. Short version of `Cases/Brexit/brexit-game.pdf`, where every step, variant and reference is worked out. Costs in £bn; payoff = −cost; players are risk neutral.

(i) Players

- **UK** (Boris Johnson's government): prepares or not; presents the EU's offer to Parliament or rejects it.
- **EU** (chief negotiator Michel Barnier): chooses which deal to offer.
- **Parliament, modelled as Nature**: passes a presented deal with probability $p = 0.8$, fails it with 0.2.

(ii) Timing and actions

Stage 1. **UK**: *Prepare* (P) or *Not prepare* (N). Preparation costs $\kappa = 5$, is all-or-nothing and sunk; it saves $m = 12.5$ of the hard-Brexit cost.

Stage 2. **EU**: offers a deal $d \in \{F, Nt, U\}$ (favourable, neutral, unfavourable to the UK).

Stage 3. **UK**: *presents* the deal to Parliament or *rejects* it (hard Brexit, plus a political cost $r \geq 0$; baseline $r = 0$).

Stage 4. **Nature**: the deal passes (0.8) or fails (0.2, hard Brexit, no r).

Strategies: $|S_{UK}| = 2 \times 2^6 = 128$, $|S_{EU}| = 3^2 = 9$.

(iii) Information

Baseline: the EU observes preparation (stockpiles and border infrastructure are visible); the UK sees the offer before deciding. Every information set is a singleton; the only uncertainty is the vote, whose probability is common knowledge. **Variant:** if the EU does not observe preparation, its two nodes form one information set, so stages 1 and 2 are effectively simultaneous (solved in the box below).

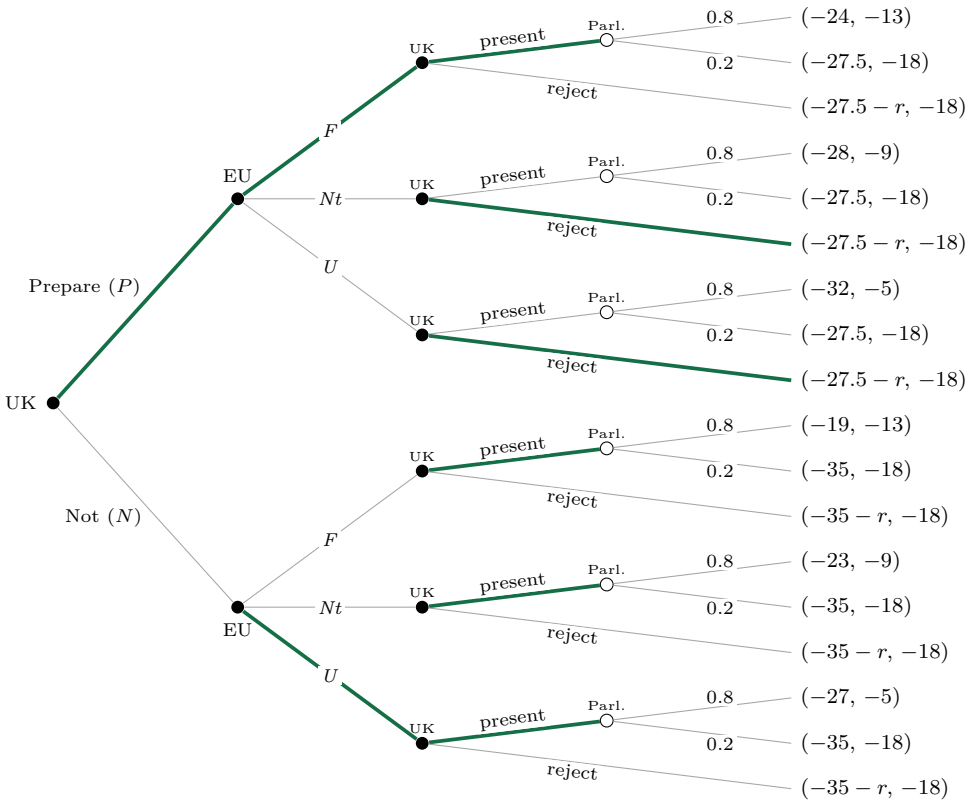
(iv) Payoffs

Inputs (costs): hard Brexit UK 35, EU 18; deals (c_d, e_d): F $(19, 13)$, Nt $(23, 9)$, U $(27, 5)$. A prepared UK's hard-Brexit cost is $35 - 12.5 + 5 = 27.5$; in a soft Brexit it pays $c_d + 5$. The 18 leaves carry 8 distinct payoff pairs (UK, EU):

UK prepared?	Event	F	Nt	U
N	deal passes	$(-19, -13)$	$(-23, -9)$	$(-27, -5)$
N	vote fails	$(-35, -18)$	$(-35, -18)$	$(-35, -18)$
N	UK rejects	$(-35 - r, -18)$	$(-35 - r, -18)$	$(-35 - r, -18)$
P	deal passes	$(-24, -13)$	$(-28, -9)$	$(-32, -5)$
P	vote fails	$(-27.5, -18)$	$(-27.5, -18)$	$(-27.5, -18)$
P	UK rejects	$(-27.5 - r, -18)$	$(-27.5 - r, -18)$	$(-27.5 - r, -18)$

Why this ordering: an unprepared UK prefers every deal to no deal ($27 < 35$); a prepared UK, net of the sunk 5, prefers no deal to Nt and U but not to F ($19 < 22.5 < 23 < 27$); the EU prefers less generous deals and any deal to no deal ($5 < 9 < 13 < 18$), and is hurt less by no deal than the UK ($18 < 35$).

(v) The game tree



Filled dots: decision nodes; open dots: Parliament (upper edge pass, 0.8; lower edge fail, 0.2). Thick green: the SPE action at every node, on and off the path, for $r < 0.4$ and $k \leq 5$.

(vi) Backward induction and the SPE

Step 1 (UK: present or reject). After N, presenting d costs $0.8\,c_d + 0.2 \times 35$: F 22.2, Nt 25.4, U 28.6, all below $35 + r$, so *present every deal*. After P, presenting costs $0.8\,(c_d + 5) + 0.2 \times 27.5$: F 24.7, Nt 27.9, U 31.1, against 27.5 for rejecting, so *present F, reject Nt and U*.

Step 2 (EU: offer). A presented deal costs the EU $0.8\,e_d + 3.6$: F 14, Nt 10.8, U 7.6; a rejection costs 18. After N everything is presented: offer U (7.6). After P only F is presented: offer F (14 beats 18).

Step 3 (UK: prepare?). P leads to $(-24.7, -14)$, N to $(-28.6, -7.6)$. Since $24.7 < 28.6$, the UK *prepares*.

SPE (baseline, $r = 0$, $k = 0$):

$$s^*_{UK} = \big(P;\ \text{after } P\text{: present } F,\ \text{reject } Nt,\ \text{reject } U;\ \text{after } N\text{: present } F, Nt, U\big),$$
$$s^*_{EU} = \big(F \text{ after } P;\ U \text{ after } N\big),$$

with Parliament passing a presented deal with probability 0.8. Path: P, F, present; payoffs $(-24, -13)$ with probability 0.8, $(-27.5, -18)$ with probability 0.2. **Expected payoffs** $(-24.7, -14)$, against $(-28.6, -7.6)$ without preparation. The UK gains 3.9, the EU loses 6.4; the 2.5 difference is a deadweight loss.

Key comparative statics

- **Political cost of walking away,** r (a prepared UK presents d iff $c_d \leq 22.5 + r/0.8$): $r < 0.4$: (P, F), UK -24.7, EU -14; $0.4 < r < 3.6$: (P, Nt), UK -27.9, EU -10.8; $r > 3.6$: (N, U), UK -28.6, EU -7.6. A higher r makes the threat costlier to carry out, so it *weakens* it.
- **EU deterrence premium** k on an implemented F (cost $13 + k$): after P the EU offers F iff $0.8\,(13 + k) + 0.2 \times 18 \leq 18$, i.e. $k \leq 5$. For $k > 5$ the prepared UK rejects and the outcome is a hard Brexit, $(-27.5, -18)$.
- **Unobservable preparation**: no pure equilibrium (best responses cycle). Mixed equilibrium $\sigma = \Pr(P) = 8/13$, $\tau = \Pr(F) = 11/36$; UK $-1199/45 \approx -26.64$ (versus -24.7 when observed), EU -14; talks break down with probability $50/117 \approx 0.43$. Hidden commitment is no commitment.

In-class decision guide (baseline numbers)

Playing the UK

- **Prepare, and make sure the EU sees it.** Visible preparation is worth 3.9 at $p = 0.8$ (and $3p+7.5(1-p) > 0$ for any p); hidden preparation is worth far less.
- **If prepared**, present d iff $c_d \leq 22.5 + r/p$: present only F while $r < 0.4$; if $0.4 < r < 3.6$, present F or Nt and reject U; if $r > 3.6$, do not prepare at all.
- **If unprepared, present whatever is offered** ($c_d \leq 35 + r/p$ always holds): every deal beats no deal, and saying otherwise is cheap talk.
- **If the EU's deterrence concern is large** ($k > 5$), expect a breakdown; you still prefer to have prepared ($27.5 < 28.6$).

Playing the EU

- **If you see preparation, offer** F, unless you believe the UK's r is high. If unsure, offer Nt instead of F only if $\Pr(r \geq 0.4) > 5/9$, and U instead of F only if $\Pr(r \geq 3.6) > 5/13$ (from $q \times 10.8 + (1-q) \times 18 = 14$ and $q \times 7.6 + (1-q) \times 18 = 14$).
- **If you see no preparation, offer** U**.** It will be presented.
- **If you cannot tell**, the equilibrium has you offering F with probability $11/36$ and U otherwise; in practice, invest in finding out.
- **Weigh precedent.** If rewarding a leaver costs you more than 5 in deterrence, refuse F and accept that a hard Brexit follows.

(vii) Why your negotiation may differ from the SPE

- **Cheap talk versus commitment.** "No deal is better than a bad deal" changes nothing unless backed by preparation; an EU that believes an unprepared UK's threat plays a Nash equilibrium that is not subgame perfect, with payoffs $(-22.2, -14)$.
- **Fairness.** Lab responders reject splits they find insulting; the surplus here is $53 - 32 = 21$, and F, Nt, U give the UK about 76%, 57%, 38% of it. An EU player may refuse F as unfair to the EU.
- **Reputation and repeated interaction.** The EU faces 27 members and future applicants (the premium k), and both sides negotiate again later.
- **Uncertainty and deadlines.** Private information about r or p can make rational bargaining break down; a rejection that leads to an extension turns the game into alternating offers, where patience matters.

Record your pair's moves against the SPE in the template at the end of §11 of `brexit-game.pdf` ("Template: record your negotiation against the SPE").